Luis Eduardo Jacobo

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Professional Summary

Mechanical engineering graduate student with experience in materials science, structural mechanics, additive manufacturing, and aerospace systems. Skilled in catalyst material synthesis, electrochemical characterization, FEA, CAD, and student leadership. Research interests include PEM fuel cell catalyst materials, advanced manufacturing, and structural analysis.

Education

Master of Science | Mechanical Engineering
Western Michigan University — Graduation: April 2027
Thesis Topic: Design and Analysis of Catalyst Materials for PEM Fuel Cells — GPA: 3.86/4.00
Kalamazoo, Michigan

Bachelor of Science | Mechanical and Aerospace Engineering
Western Michigan University | Lee Honors College — Graduation: April 2025
Concentration: Solid Mechanics and Structures — GPA: 3.83/4.00
Kalamazoo, Michigan — Magna Cum Laude

Bachelor of Science | Mechanical Engineering
Instituto Tecnológico de Santo Domingo — Graduation: October 2025
Focus Area: Medical Device Manufacturing, Industry Standards, and Metrology — GPA: 3.88/4.00
Santo Domingo, Dominican Republic — Summa Cum Laude

Research Experience

Researcher: Carbon Materials for PEM Fuel Cell Catalysts — April 2024 – Present
Western Michigan University

- Design and synthesize polyoxometalate-doped carbon catalyst materials for PEM fuel cells.
- Perform physicochemical and electrochemical characterization using FTIR, BET, SEM, Raman, XRD, CV, and LSV.
- Establish structure–property relationships between microstructure, surface chemistry, and catalytic performance.

Materials Scientist & Engineer — May 2024 – December 2025
CURIE-S / NASA L'SPACE Program

- Lead materials, manufacturing, and design work for a NASA-funded proposal securing $10,000 in funding.
- Contributed TX12 expertise in materials, structures, manufacturing, ceramics, chemistry, and additive manufacturing.
- Supported proposal development, technical research, and Siemens NX-based design work.

Research Assistant: Dominican QSat — January 2023 – July 2023
Instituto Tecnologico de Santo Domingo

- Identified on-board processing methods to be implemented on the Dominican QSat.
- Investigated satellite studies and imaging used in Algae and Sargassum.
- Reported summarized findings to the Aerospace Science office.

Technical Skills

- Laboratory & Characterization: FTIR, BET, SEM, Raman, XRD, CV, LSV, electrochemical testing, mechanical testing
- Programming Languages: Python, MATLAB
- Engineering & Simulation: Abaqus CAE, Siemens NX, SolidWorks, AutoCAD, GMAT, LabVIEW, Granta EduPack
- Manufacturing & Design: Ceramic SLA, 3D printing, laser cutting, welding, soldering, CAD, technical drawing, metrology
- Standards & Regulations: ASTM, ISO 17025, CFR 820
- Productivity & Digital Tools: Microsoft Office, digital database management, digital literacy
- Languages: Spanish (Fluent), English (Fluent), French (B1), German (Basic)

Publications & Presentations

- *Jacobo Mejia, L., et al.* "Synthesis and Characterization of Carbon-Based Catalyst Materials with Molybdenum Polyoxometalates." Poster presented at the 10th International Conference on Fundamentals & Developments of Fuel Cells and Electrolyzers, 2026.

- Jacobo Mejia, Luis, "Synthesis and Characterization of Polyoxometalate-doped Carbon Material for Proton Exchange Membrane (PEM) Fuel Cells" (2025). *Honors Theses*. 3927. https://scholarworks.wmich.edu/honors_theses/3927

- *Jacobo Mejia, L., et al.* Optimization of Synthesis Parameters for Rapid Mass-Production of Graphdiyne Using Bipolar Electrochemistry. Manuscript in preparation.

- *Jacobo Mejia, L., Klang, G., et al.* Preparation of a Discharge Channel for a Krypton-Powered Hall Effect Thruster (HET) Using Ceramic Stereolithography. Manuscript in preparation.

Professional Experience

Teaching Assistant — August 2025 – Present

College of Engineering and Applied Sciences - WMU

- Assisted a class of 60 students with understanding advanced Structural Mechanics concepts.
- Guided students with Finite Element Analysis (FEA) projects using Abaqus.
- Graded weekly homework assignments and helped create problems.

Graduate Student Ambassador — August 2024 – Present

Graduate College - WMU

- Represented the graduate college at open house and promotional events.
- Gave tours and answered questions for prospective graduate students.
- Met with deans, department chairs, and academic advisors to discuss programs.

Teaching Assistant — August 2022 – July 2023

Instituto Tecnologico de Santo Domingo

- Assisted students in mastering computer aided design (CAD) software.
- Explained concepts regarding technical drawing, geometric dimensioning and tolerancing.
- Coached students in preparation for CSWA and CSWP exams.

Operation and Logistics Intern — November 2022 – January 2023

Instituto Tecnologico de Santo Domingo

- Cataloged extensive documentation in a digital database and on a physical location.
- Created a digital database to facilitate access, visualization, and location of files.
- Maintained confidentiality standards under NDA guidelines.

Academic Projects & Technical Work

- **Variational Analysis of 3D-Printed Wing Sections:** Conducted Abaqus CAE vibration analysis of 3D-printed wing sections with varying infill patterns and percentages.

- **3D Printed Aircraft Competition:** Helped design a 1-meter-wingspan PLA aircraft for a national 3D-printed aircraft competition.

- **FDM Structural Analysis with NASMAT as Delamination Predictor:** Developed a Python/FEA-based method to predict FDM delamination from g-code using NASMAT and the Generalized Method of Cells.

- **FEA of a Fuselage Section with Abaqus:** Modeled a commercial aircraft fuselage section in Abaqus CAE, simulated loading conditions, and prepared a structural analysis and optimization report.

- **Mechanical Properties of PLA:** Tested tensile strength, flexural modulus, and impact yield of 3D-printed PLA across different print speeds and orientations under ASTM and ISO 17025 guidelines.

- **Truss Structure:** Designed, manufactured, and analyzed a CAD truss structure to meet a target failure load within a narrow tolerance range.

Leadership & Academic Service

Graduate Student Association — April 2026 – Present

President

- Represent graduate student interests in meetings with university leadership, faculty, and administrative offices.
- Lead executive board initiatives focused on graduate student advocacy, engagement, and campus-wide support.
- Manage organizational priorities, communication, and funded initiatives serving the graduate student body.

Solid Mechanics and Structures Hub (SMASH) — April 2024 – Present

Co-Founder, Vice President, and R&D Engineer

- Founded a registered student organization at Western Michigan University.
- Secure yearly funding for 3D printing machines, materials, parts, tools, and other resources supporting student-led engineering projects.
- Apply mechanical design, structural mechanics, and 3D printing skills to support aircraft design and optimization for the 7th Annual 3D Printed Aircraft Competition.

Western Student Association — April 2025 – May 2026

Director of Allocations

- Oversaw the fair allocation of $1,000,000 in student funds to student organizations.
- Created and updated SOPs and bylaws for the allocation of student funds.
- Chaired the Allocations Commission and served in the executive branch of student government.

Western Student Association — November 2024 – April 2025

Academic Chair

- Participated in Student Body Assembly meetings and student government initiatives.
- Represented the Graduate College within student government and advocated for graduate student concerns.
- Recognized by the Speaker of the Assembly for contributions to student government and campus leadership.

Society of Hispanic Professional Engineers — February 2024 – April 2025

President

- Led a registered student organization with over 60 active members.
- Networked with RSOs, companies, and university partners to secure over $21,000 in funding.
- Encouraged professional development through workshops, networking activities, and mentorship.

Western Aerospace Launch Initiative (WALI) — August 2023 – April 2025

Team Member and Mechanical Designer

- Created, optimized and tested the CAD model for the CanSat mission.
- Member of the structural subsystem for the CubeSat mission
- Member of the thermal subsystem for the CubeSat mission

Professional Certifications

SolidWorks Associate – Additive Manufacturing	February 2024
SolidWorks Professional – Mechanical Design	October 2023
SolidWorks Associate – Mechanical Design	August 2022
Autodesk AutoCAD Certified User	August 2022
Python Essentials 1	January 2023

Professional Memberships

Tau Beta Pi – Kappa Chapter	November 2024 - Present
American Society of Mechanical Engineers (WMU Chapter)	September 2023 – Present
Society of Hispanic Professional Engineers	August 2023 – Present

Honors and Awards

- Stapleton-Reinhold Scholarship — August 2025
- Global Education Merit Scholarship — August 2025
- WMU Student Sustainability Grant — February 2025
- Lee Honors College Research and Creative Scholarship — November 2024
- ScholarSHPE Undergraduate Scholarship — July 2024
- Mechanical and Aerospace Merit Scholarship — April 2024
- Dean's List – College of Engineering and Applied Sciences — December 2023
- Iftikhar Kamil Madni Science and Engineering Memorial Scholarship — July 2023
- Global Education Merit Scholarship — July 2023
- MESCyT National Scholarship / Beca Nacional MESCyT — August 2021
- Microsoft Office Specialist (MOS) National Champion — May 2019

Volunteer Experience

Aircraft Restoration Volunteer — June 2024 – August 2024
Air Zoo Aerospace & Science Center

- Restored historic aircraft in support of Navy preservation efforts.
- Assisted with sheet metal fabrication, woodworking, and period-accurate finishing.
- Collaborated with engineers and restoration experts to ensure historical accuracy and project quality.

Food Pantry Assistant — May 2024 – August 2025
Montgomery Essential Needs Program

- Sorted, organized, and distributed food donations to community members in need, serving an average of 50 students per week.
- Maintained a clean and efficient food storage area in compliance with food safety guidelines.
- Provided compassionate customer service and connected community members with additional local resources.

Part-Time Volunteer — November 2013 – January 2021
Limesa Motors

- Gained hands-on experience with vehicle maintenance, importation, and fabrication.
- Supported daily operations across multiple areas of the business as needed.
- Assisted with onboarding and training of new team members.